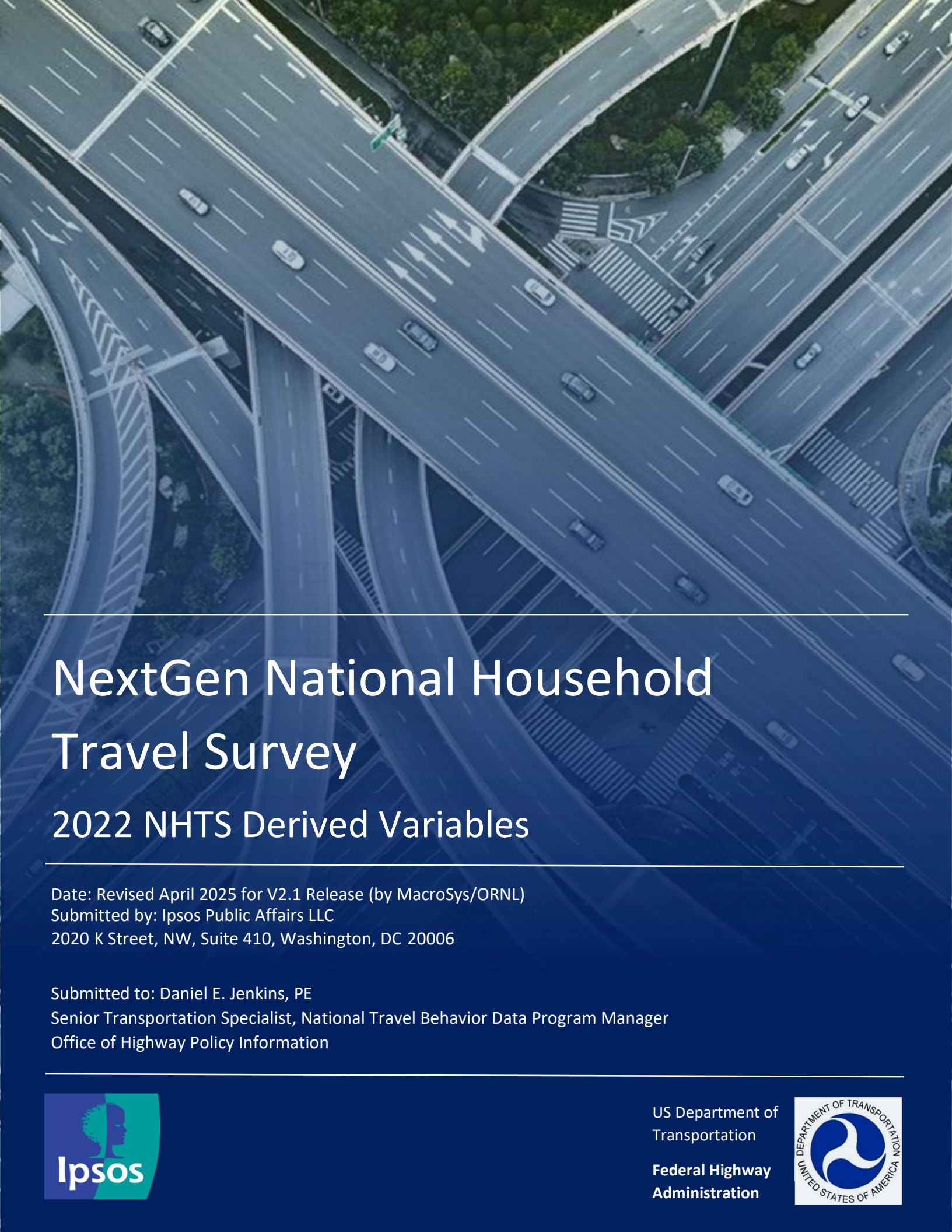




NextGen National Household Travel Survey

2022 NHTS Derived Variables

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US Department of
Transportation

Federal Highway
Administration





Derived Variables

Variable and Creation Instructions	Short Description and Ranges
AIRSIZE Creation: Use airport size related to domestic airport (AIRCODE)	Description: Domestic airport hub size based on FY23 NPIAS hub type status 01=Nonhub 02=Small 03=Medium 04=Large
ANNMILES Creation: If VEHOWNED=-7 or -8 or -9 OR VEHMILES=-7 or -8 or -77 or -88 or -9 or ESTMILES=-7 or -8 or -77 or -88 or -9 then ANNMILES=-9 Else If VEHOWNED=1 then ANNMILES=VEHMILES; /*This should result in positive values and values of -9*/ Else if VEHOWNED=2 then ANNMILES= (ESTMILES/VEHOWNEDMO)*12; If VEHOWNED=2 and VEHOWNMO=-7 or -8 or -9 then ANNMILES=-9 If ANNMILES > 200000 then ANNMILES=200000	Description: Self-reported annualized mile estimate. Ranges: Continuous variable
BEGTRIP Creation: Remove day of month information from LD5 (YYYYMMDD)	Description: Beginning date of trip (YYYYMM)
CDIVMSAR Creation:	Description: Grouping of household by combination of Census division, MSA status, and presence of rail Range:

Variable and Creation Instructions	Short Description and Ranges
If CENSUS_D=1 then do; If MSACAT=1 then CDIVMSAR=11; If MSACAT=2 then CDIVMSAR=12; If MSACAT=3 then CDIVMSAR=13; If MSACAT=4 then CDIVMSAR=14; end; If CENSUS_D=2 then do; If MSACAT=1 then CDIVMSAR=21; If MSACAT=2 then CDIVMSAR=22; If MSACAT=3 then CDIVMSAR=23; If MSACAT=4 then CDIVMSAR=24; end; If CENSUS_D=3 then do; If MSACAT=1 then CDIVMSAR=31; If MSACAT=2 then CDIVMSAR=32; If MSACAT=3 then CDIVMSAR=33; If MSACAT=4 then CDIVMSAR=34; end; If CENSUS_D=4 then do; If MSACAT=1 then CDIVMSAR=41; If MSACAT=2 then CDIVMSAR=42; If MSACAT=3 then CDIVMSAR=43; If MSACAT=4 then CDIVMSAR=44; end; If CENSUS_D=5 then do; If MSACAT=1 then CDIVMSAR=51; If MSACAT=2 then CDIVMSAR=52; If MSACAT=3 then CDIVMSAR=53; If MSACAT=4 then CDIVMSAR=54; end; If CENSUS_D=6 then do; If MSACAT=1 then CDIVMSAR=61; If MSACAT=2 then CDIVMSAR=62; If MSACAT=3 then CDIVMSAR=63; If MSACAT=4 then CDIVMSAR=64; end; If CENSUS_D=7 then do; If MSACAT=1 then CDIVMSAR=71; If MSACAT=2 then CDIVMSAR=72; If MSACAT=3 then CDIVMSAR=73; If MSACAT=4 then CDIVMSAR=74;	11=New England (ME, NH, VT, CT, MA, RI) MSA or CMSA of 1 million or more with heavy rail 12=New England (ME, NH, VT, CT, MA, RI) MSA or CMSA of 1 million or more without heavy rail 13=New England (ME, NH, VT, CT, MA, RI) MSA of less than 1 million 14=New England (ME, NH, VT, CT, MA, RI) Not in a MSA 21=Mid-Atlantic (NY, NJ, PA) MSA or CMSA of 1 million or more with heavy rail 22=Mid-Atlantic (NY, NJ, PA) MSA or CMSA of 1 million or more without heavy rail 23=Mid-Atlantic (NY, NJ, PA) MSA of less than 1 million 24=Mid-Atlantic (NY, NJ, PA) Not in a MSA 31=East North Central (IL, IN, MI, OH, WI) MSA or CMSA of 1 million or more with heavy rail 32=East North Central (IL, IN, MI, OH, WI) MSA or CMSA of 1 million or more without heavy rail 33=East North Central (IL, IN, MI, OH, WI) MSA of less than 1 million 34=East North Central (IL, IN, MI, OH, WI) Not in a MSA 41=West North Central (IA, KS, MO, MN, ND, NE, SD) MSA or CMSA of 1 million or more with heavy rail 42=West North Central (IA, KS, MO, MN, ND, NE, SD) MSA or CMSA of 1 million or more without heavy rail 43=West North Central (IA, KS, MO, MN, ND, NE, SD) MSA of less than 1 million 44=West North Central (IA, KS, MO, MN, ND, NE, SD) Not in a MSA 51=South Atlantic (DC, DE, FL, GA, MD, NC, SC, WV, VA) MSA or CMSA of 1 million or more with heavy rail 52=South Atlantic (DC, DE, FL, GA, MD, NC,

Variable and Creation Instructions	Short Description and Ranges
end; If CENSUS_D=8 then do; If MSACAT=1 then CDIVMSAR=81; If MSACAT=2 then CDIVMSAR=82; If MSACAT=3 then CDIVMSAR=83; If MSACAT=4 then CDIVMSAR=84; end; If CENSUS_D=9 then do; If MSACAT=1 then CDIVMSAR=91; If MSACAT=2 then CDIVMSAR=92; If MSACAT=3 then CDIVMSAR=93; If MSACAT=4 then CDIVMSAR=94; end;	SC, WV, VA) MSA or CMSA of 1 million or more without heavy rail 53=South Atlantic (DC, DE, FL, GA, MD, NC, SC, WV, VA) MSA of less than 1 million 54=South Atlantic (DC, DE, FL, GA, MD, NC, SC, WV, VA) Not in a MSA 62=East South Central (AL, KY, MS, TN) MSA or CMSA of 1 million or more without heavy rail 63=East South Central (AL, KY, MS, TN) MSA of less than 1 million 64=East South Central (AL, KY, MS, TN) Not in a MSA 71=West South Central (AR, LA, OK, TX) MSA or CMSA of 1 million or more with heavy rail 72=West South Central (AR, LA, OK, TX) MSA or CMSA of 1 million or more without heavy rail 73=West South Central (AR, LA, OK, TX) MSA of less than 1 million 74=West South Central (AR, LA, OK, TX) Not in a MSA 81=Mountain (AZ, CO, ID, MT, NM, NV, UT, WY) MSA or CMSA of 1 million or more with heavy rail 82=Mountain (AZ, CO, ID, MT, NM, NV, UT, WY) MSA or CMSA of 1 million or more without heavy rail 83=Mountain (AZ, CO, ID, MT, NM, NV, UT, WY) MSA of less than 1 million 84=Mountain (AZ, CO, ID, MT, NM, NV, UT, WY) Not in a MSA 91=Pacific (AK, CA, HI, OR, WA) MSA or CMSA of 1 million or more with heavy rail 92=Pacific (AK, CA, HI, OR, WA) MSA or CMSA of 1 million or more without heavy rail 93=Pacific (AK, CA, HI, OR, WA) MSA of less than 1 million

Variable and Creation Instructions	Short Description and Ranges
	94=Pacific (AK, CA, HI, OR, WA) Not in a MSA
CENSUS_D Creation: Based on CONFIRMEDHOME_STATE; 1= ME, NH, VT, CT, MA, RI; 2= NY, NJ, PA; 3=IL, IN, MI, OH, WI; 4= IA, KS, MO, MN, ND, NE, SD 5= DC, DE, FL, GA, MD, NC, SC, WV, VA; 6= AL, KY, MS, TN; 7= AR, LA, OK, TX; 8= AZ, CO, ID, MT, NM, NV, UT, WY; 9= AK, CA, HI, OR, WA	Description: Census division classification for the respondent's confirmed home address. Range: 01=New England 02=Middle Atlantic 03=East North Central 04=West North Central 05=South Atlantic 06=East South Central 07=West South Central 08=Mountain 09=Pacific
CENSUS_R Creation: If CENSUS_D in (1,2) then CENSUS_R=1; If CENSUS_D in (3,4,) then CENSUS_R=2; If CENSUS_D in (5,6,7) then CENSUS_R=3; If CENSUS_D in (8,9) then CENSUS_R=4;	Description: Census region classification for confirmed home address. Range: 01=Northeast 02=Midwest 03=South 04=West
CNTTDHH Creation: CNTTDHH= the COUNT of trip records within each HOUSEID from the TRIPS file contained in the HOUSEHOLD file.	Description: Total number of trips for all household members ages 5+on travel day. Range: 0-45
CNTTDTR	Description:

Variable and Creation Instructions	Short Description and Ranges
Creation: If B1A=2 then CNTTDTR=0 Otherwise CNTTDTR = the total number of trips a person reported. COUNT of trip records within PERSONID from the TRIPS file contained in the PERSON file.	Total number of trips a person entered in their travel record on travel day. Range: 0-36
DRIVER Creation: If R_AGE>0 and R_AGE<15 then DRIVER=-1; Else If R_AGE>=15 and DRVR = 1 THEN DRIVER=01; Else IF COUNT(TRIPID WHERE WHODROVE = PERSONID) > 0 THEN DRIVER=01; Else IF COUNT(TRIPID WHERE WHODROVE = PERSONID) = 0 THEN DRIVER=02;	Description: Driver status, derived. Range: 01= Yes 02= No -1=Valid skip
DRVR_FLG Creation: If (TRPTRANS equal to 1, 2, 3, 4, 6, or 7) and WHODROVE_IMP=PERSONID then DRVR_FLG=01; If (TRPTRANS equal to 1, 2, 3, 4, 6, or 7) and WHODROVE_IMP NOT= PERSONID then DRVR_FLG=02; If (TRPTRANS NOT equal to 1, 2, 3, 4, 6, or 7) then DRVR_FLG=-1;	Description: Flag for driver on trip. Range: 01= Driver on trip 02= Not driver on trip -1= Valid skip -9= Not ascertained
DRVRCNT Creation: COUNT of persons in HH where DRIVER=01.	Description: Number of drivers in the household. Range: 0-7
DWELTIME	Description: Time at Destination in minutes.

Variable and Creation Instructions	Short Description and Ranges
Creation: Difference between ENDTIME_local_24 from current trip record and STARTIME_local_24 from next trip record, in minutes. The final trip of the travel day is assigned a value of -9, as are any trips where start or end times are unreported	Range: Continuous variable
ENDTRIP Creation: Remove day of month information from LD6 (YYYYMMDD)	Description: Ending date of trip (YYYYMM)
EXITCDIV Creation: Assign FARST value to respective division. See CENSUS_D for Census division-state crosswalk.	Description: Census division at which respondent exited the US (international trips only)
FARCDIV Creation: Assign FARST value to respective division. See CENSUS_D for Census division-state crosswalk.	Description: Farthest domestic destination: Census Division
FARCREG Creation: Assign FARST value to respective region. See CENSUS_R for Census division-state crosswalk.	Description: Farthest domestic destination: Census region
FLAG100 Creation: If (Count of person records within the household where PFLAG=1/ELIGDIARIES)=1 then FLAG100=1;	Description: Flag to indicate proportion of household members ages 5+ who completed the survey. Range:

Variable and Creation Instructions	Short Description and Ranges
Else if 1> (Count of person records within the household where PFLAG=1/ELIGDIARIES)>=.75 then Flag100=2;	1= All eligible household members completed 2= 75% to 99% of eligible household members completed
GASPRICE Creation: Assign PADD Region based on HHSTATE (see GASPRICE tab). Match record to gas price for region found at https://www.eia.gov/petroleum/gasdiesel/xls/pswrgvwall.xls based upon PADD region and diary_date.	Description: Weekly regional gasoline price based upon "all grades conventional" during the week of the household's travel day, in cents. Range: 272.7-597.9
GCD_FLAG Creation: If GCDTOT=-1,-9 then GCD_FLAG=GCDTOT Else If GCDTOT > 49.5 then GCD_FLAG=01 Else GCD_FLAG=02	Description: Flag for long distance trips of 50 miles or more
GCDTOT Creation: Based on CONFIRMEDHOME_FormattedAddr and LD3_CITY Minimum geodesic (Great Circle) distance between home location and long distance destination in miles, using WGS84 coordinate system.	Description: Minimum geodesic (Great Circle) distance between home location and long distance destination location in miles. Range: 0-5005 -9= Not ascertained
GCDWORK Creation: Based on WORKADDRESS_FormattedAddr and CONFIRMEDHOME_FormattedAddr where	Description: Minimum geodesic (Great Circle) distance between home location and work location in miles. Range:

Variable and Creation Instructions	Short Description and Ranges
EMPLOYED = 1 and WORKLOC = 1 OR WORKLOC = 2 AND WKFHMHM20 = 1 OR WKFHMHM20 = 2 OR WKFHMHM20 = 3 Minimum geodesic (Great Circle) distance between home location and work location in miles, using WGS84 coordinate system. https://geographiclib.sourceforge.io/html/C/geodesic_8h.html#a19bc3d000428010ad9d8509174e672c9	0-2544 -9= Not ascertained
HH_HISP Creation: HH_HISP=R_HISP for primary household respondent (PERSONID=01);	Description: Hispanic status of primary (household) respondent. Range: 01= Hispanic or Latino 02= Not Hispanic or Latino
HH_RACE Creation: HH_RACE=R_RACE for primary household respondent (PERSONID=01);	Description: Race of household respondent. Range: 01=White 02=Black or African American 03=Asian 04=American Indian/Alaska Native 05=Native Hawaiian/Pacific Islander 06=Multiple races selected 97=Other race
HHFAMINC_IMP Creation: Income of subject used in weighting. Replace values in HHFAMINC_IMP that are -7 or -8 with the imputed income values.	Description: Household income (imputed). Range: 01= Less than \$10,000

Variable and Creation Instructions	Short Description and Ranges
	02= \$10,000 to \$14,999 03= \$15,000 to \$24,999 04= \$25,000 to \$34,999 05= \$35,000 to \$49,999 06= \$50,000 to \$74,999 07= \$75,000 to \$99,999 08= \$100,000 to \$124,999 09= \$125,000 to \$149,999 10= \$150,000 to \$199,999 11= \$200,000 or more
HHMEMDRV Creation: IF TRPTRANS is not in (1,2,3,4,5,6,7) THEN -1 IF WHODROVE_IMP is not = 97 AND TRPTRANS = (1,2,3,4,5,6,7) THEN 1 IF WHODROVE_IMP = 97 AND TRPTRANS = (1,2,3,4,5,6,7) THEN 2	Description: Household member drove on trip. Range: 1= Household member drove 2= Non-household member drove
HHRELATD Creation: If HHSIZE=1, then HHRELATD=2; Else if the COUNT of records on the PERSON file for the household where R_RELAT is in (1, 2, 3, 4, 5) >0 then HHRELATD=1; Else HHRELATD=2	Description: Flag indicating at least 2 persons in HH are related. Range: 01= At least two persons in HH are related 02= No related persons in HH
HHSIZE Creation: HHSIZE = Sum (of NUMADULT, PPT517, YOUNGCHILD)	Description: Total number of people in household. Range:

Variable and Creation Instructions	Short Description and Ranges
	01 - 10
HHVEHCNT Creation: Sum of records with same HOUSEID in vehicle file	Description: Total number of vehicles in household. Range: Range is zero to X, X being the highest number of vehicles per household 0 - 17
HYBRID Creation: If VEHFUEL in ("04", "05", "06") then HYBRID=01 Else HYBRID=02	Description: Hybrid vehicle. Range: 01= Yes 02= No
LDT_FLAG Creation: If data taken from trip file then LDT_FLAG=01 Else if PFLAG=02 then LD_FLAG=-1 Else if LD questions not asked LDT_FLAG=-9 Else if LD4_5YAGO=97 then LDT_FLAG=03 Else if LD4_5YAGO=98 then LDT_FLAG=04 Else if LD4 not -1,-9 or LD6 not -1,-9 then LDT_FLAG=02 Else if MRT_DATE not -1,-9 then LDT_FLAG=02 Else if LD3_CITY not -1,-9 then LDT_FLAG=02 Else if LONGDIST > 0 then LDT_FLAG=02 Else if LD7 > 0 then LDT_FLAG=02 Else LDT_FLAG=-9	Description: Source of long distance data
LIF_CYC	Description:

Variable and Creation Instructions	Short Description and Ranges
Creation: IF R_AGE < 18 THEN "CHILD" IF R_AGE > 21 THEN "ADULT" IF R_AGE BETWEEN 18 AND 21 THEN IF R_RELAT = 2 THEN "CHILD" IF R_RELAT = (-7,-8,-9,4,5,7) AND [ANY OTHER HH MEMBER] R_RELAT = 3 THEN "CHILD" IF R_RELAT = ((-7,-8,-9,4,5,7) AND NOT = [ANY OTHER HH MEMBER] R_RELAT = 3 THEN "ADULT" IF R_RELAT = 6 AND [ANY OTHER HH MEMBER] (R_RELAT = 1 AND R_AGE > 21) THEN "CHILD" IF R_RELAT = 6 AND NOT = [ANY OTHER HH MEMBER] (R_RELAT = 1 AND R_AGE > 21) THEN "ADULT" IF R_RELAT = (1,3) THEN "ADULT" IF HHSIZE=1 THEN "ADULT" HOUSEHOLD_ADULT_COUNT = COUNT (WHERE ADULT_CHILD_STATUS = "ADULT") HOUSEHOLD_CHILD_COUNT = COUNT (WHERE ADULT_CHILD_STATUS = "CHILD") HOUSEHOLD_RETIRED_COUNT = COUNT (WHERE PRMACT = 5 OR ((PRMACT IS MISSING OR PRMACT = (-7,-8)) AND AGE >= 65)) MIN_AGE = MINIMUM (R_AGE) for HOUSEHOLD LIF_CYC = IF HOUSEHOLD_ADULT_COUNT = 1 AND HOUSEHOLD_CHILD_COUNT = 0 AND HOUSEHOLD_RETIRED_COUNT = 0 THEN 01 IF HOUSEHOLD_ADULT_COUNT >= 2 AND HOUSEHOLD_CHILD_COUNT = 0 AND HOUSEHOLD_RETIRED_COUNT = 0 THEN 02 IF HOUSEHOLD_ADULT_COUNT = 1 AND HOUSEHOLD_CHILD_COUNT >= 1 AND MIN_AGE BETWEEN 0 AND 5 THEN 03	Life Cycle classification for the household. Range: 01=one adult, no children 02=2+ adults, no children 03=one adult, youngest child 0-5 04=2+ adults, youngest child 0-5 05=one adult, youngest child 6-15 06=2+ adults, youngest child 6-15 07=one adult, youngest child 16-21 08=2+ adults, youngest child 16-21 09=one adult, retired, no children 10=2+ adults, retired, no children

Variable and Creation Instructions	Short Description and Ranges
IF HOUSEHOLD_ADULT_COUNT >= 2 AND HOUSEHOLD_CHILD_COUNT >= 1 AND MIN_AGE BETWEEN 0 AND 5 THEN 04 IF HOUSEHOLD_ADULT_COUNT = 1 AND HOUSEHOLD_CHILD_COUNT >= 1 AND MIN_AGE BETWEEN 6 AND 15 THEN 05 IF HOUSEHOLD_ADULT_COUNT >= 2 AND HOUSEHOLD_CHILD_COUNT >= 1 AND MIN_AGE BETWEEN 6 AND 15 THEN 06 IF HOUSEHOLD_ADULT_COUNT = 1 AND HOUSEHOLD_CHILD_COUNT >= 1 AND MIN_AGE BETWEEN 16 AND 21 THEN 07 IF HOUSEHOLD_ADULT_COUNT >= 2 AND HOUSEHOLD_CHILD_COUNT >= 1 AND MIN_AGE BETWEEN 16 AND 21 THEN 08 IF HOUSEHOLD_ADULT_COUNT = 1 AND HOUSEHOLD_CHILD_COUNT = 0 AND HOUSEHOLD_RETIRED_COUNT = 1 THEN 09 IF HOUSEHOLD_ADULT_COUNT >= 2 AND HOUSEHOLD_CHILD_COUNT = 0 AND HOUSEHOLD_RETIRED_COUNT >= 1 THEN 10	
LOOP_TRIP Creation: IF DESTINATION_LOCCODE=997 then LOOP_TRIP = 01; ELSE LOOP_TRIP = 02	Description: Trip origin and destination at same location. Range: 01=Not a loop trip 02= Loop trip
MRT_DATE Creation: Derive from LD4_YEAR,LD4_MONTH Then	Description: Date of most recent long distance trip (YYYYMM)

Variable and Creation Instructions	Short Description and Ranges
If LD4_5YAGO in 97,98 then MRT_DATE=-1 Else if LD4_MONTH in -1,-9 then MRT_DATE=LD4_MONTH If ENDTRIP not -1,-9 then MRT_DATE=ENDTRIP Else if BEGTRIP not -1,-9 then MRT_DATE=BEGTRIP Else if ENDTRIP=-9 and BEGTRIP=-9 then MRT_DATE=-9	
MSACAT Creation: IF MSASIZE = (04,05) THEN IF RAIL = 1 THEN 01 IF RAIL = 2 THEN 02 IF MSASIZE = (01,02,03) THEN 03 IF CBSA IS MISSING THEN 04	Description: Metropolitan Statistical Area (MSA) category for the household's home address, based on household's confirmed home geocode and 2020 TIGER/Line Shapefiles. Range: 01=MSA of 1 million or more, with rail 02=MSA of 1 million or more, no rail 03=MSA less than 1 million 04=Not in MSA
MSASIZE Creation: IF [POPULATION OF MSA] < 250,000 THEN "01" IF [POPULATION OF MSA] >= 250,000 AND <= 499,999 THEN "02" IF [POPULATION OF MSA] >= 500,000 AND <= 999,999 THEN "03" IF [POPULATION OF MSA] >= 1,000,000 AND <= 2,999,999 THEN "04" IF [POPULATION OF MSA] >= 3,000,000 THEN "05" IF MSA IS MISSING THEN "06"	Description: Population size category of the MSA for the confirmed home address, from the five-year ACS API. Range: 01=In an MSA of Less than 250,000 02=In an MSA of 250,000 - 499,999 03=In an MSA of 500,000 - 999,999 04=In an MSA or CMSA of 1,000,000 - 2,999,999 05=In an MSA or CMSA of 3 million or more 06=Not in MSA or CMSA
NTSAWAY	Description:

Variable and Creation Instructions	Short Description and Ranges
Creation: LD6-LD5	Nights away on long distance trip
OUTOFTOWN Creation: If B1A=02 or FRSTHM=01 or COUNT(TRIPID where LOCTYPE=01) > 0 or GCD between TRIP END and HOME < 50 then OUTOFTWN=02; Else OUTOFTWN=01;	Description: Away from home entire travel day. Range: 01=Yes 02=No
PSGR_FLG Creation: If (TRPTRANS equal to 1, 2, 3, 4, 6, or 7) and WHODROVE_IMP NOT= PERSONID then PSGR_FLG=01; If (TRPTRANS equal to 1, 2, 3, 4, 6, or 7) and WHODROVE_IMP = PERSONID then PSGR_FLG=02; If (TRPTRANS NOT equal to 1, 2, 3, 4, 6, or 7) then PSGR_FLG=-1;	Description: Flag for passenger on trip Range: 01= Passenger on trip 02= Not passenger on trip -1= Valid skip
PUBTRANS Creation: If TRPTRANS in (8, 10, 11, 12) then PUBTRANS=01; Else PUBTRANS=02;	Description: Used public transit on trip. Ranges: 01= Used public transit 02= Did not use public transit
R_RACE Creation: If SUM(RACE_1-RACE_6, RACE_SE) >1 then R_RACE='06'; Else if SUM (RACE_1-RACE_6, RACE_SE)=1 then do; If RACE_1>0 then R_RACE='01';	Description: Collapsed race. Range: 01=White 02=Black or African American

Variable and Creation Instructions	Short Description and Ranges
if RACE_2>0 then R_RACE='02'; if RACE_3>0 t then R_RACE='03'; if RACE_4>0 then R_RACE='04'; if RACE_5>0 then R_RACE='05'; if RACE_SE>0 then R_RACE='97'; If R_RACE is still blank, set R_RACE to -9;	03=Asian 04=American Indian/Alaska Native 05=Native Hawaiian/Pacific Islander 06=Multiple races selected 97=Other race -9 = Not ascertained
R_RACE_IMP Creation: If R_RACE ne -9, set R_RACE_IMP to R_RACE; If R_RACE=-9 then impute R_RACE_IMP	Description: Imputed race. Range: 01=White 02=Black or African American 03=Asian 04=American Indian/Alaska Native 05=Native Hawaiian/Pacific Islander 06=Multiple races selected 97=Other race
R_SEX_IMP Creation: Sex of subject used in weighting. Replace values in R_SEX that are -7, -8 or -9 with the imputed sex values.	Description: Sex (imputed).
RESP_CNT Creation: Count of Persons in HH where PFLAG = 01; Count (PERSONID where PFLAG = 01) in HH	Description: Count of responding persons in household. Range: 0-10
TDCASEID	Description:

Variable and Creation Instructions	Short Description and Ranges
Creation: TDCASEID equals a concatenation of HOUSEID PERSONID TRIPID	Unique ID for every trip record in trip file. Range: 90000130020101-90002180400103
TDAYDATE Creation: Extract of YYYYMM from Diary_Date	Description: Date of travel day (YYYYMM). Range: 202201-202301
TDWKND Creation: IF [TRAVDAY] = (1,7) OR ([TRAVDAY] = (6) AND STRTIME_LOCAL_24 >= 1800) THEN TRAVDAY=01; ELSE TRAVDAY=02	Description: Weekend trip. Range: 01= Weekend 02= Weekday
TRAVDAY Creation: if day_of_week=00 then TRAVDAY=01; if day_of_week=01 then TRAVDAY=02; if day_of_week=02 then TRAVDAY=03; if day_of_week=03 then TRAVDAY=04; if day_of_week=04 then TRAVDAY=05; if day_of_week=05 then TRAVDAY=06; if day_of_week=06 then TRAVDAY=07;	Description: Travel day – day of week. Range: 01= Sunday 02= Monday 03= Tuesday 04= Wednesday 05= Thursday 06= Friday 07= Saturday
TRIPMODE	Description: General mode of trip.

Variable and Creation Instructions	Short Description and Ranges
Creation: IF TRPTRANS=(01,02,03,04,05,06,07) AND DRVR_FLG=01 THEN TRIPMODE=01 ELSE IF TRPTRANS=(01,02,03,04,05,06,07) AND PSGR_FLG=01 THEN TRIPMODE=02 ELSE IF TRPTRANS=(08,10,11,12) THEN TRIPMODE=03 ELSE IF TRPTRANS=09 THEN TRIPMODE=04 ELSE IF TRPTRANS=20 THEN TRIPMODE=05 ELSE IF TRPTRANS=18 THEN TRIPMODE=06 ELSE TRIPMODE=07	Range: 01=Privately Owned Vehicle-Driver 02=Privately Owned Vehicle-Passenger 03=Public Transit 04=School Bus 05=Walk 06=Bike 07=Other
TRIPPURP Creation: IF WHYFROM = -9 OR WHYTO = -9 THEN TRIPPURP= -9 Else IF WHYFROM = (1,2) AND WHYTO = (3,4,5) THEN TRIPPURP= 01 Else IF WHYFROM = (3,4,5) AND WHYTO = (1,2) THEN 01 Else IF WHYFROM = (1,2) AND WHYTO = (12,13) THEN 02 Else IF WHYFROM = (12,13) AND WHYTO = (1,2) THEN 02 Else IF WHYFROM = (1,2) AND WHYTO = (15,16,17,18) THEN 03 Else IF WHYFROM = (15,16,17,18) AND WHYTO = (1,2) THEN 03 Else IF WHYFROM = (1,2) AND WHYTO = (6,7,8,9,10,11,14,19,97) THEN 04 Else IF WHYFROM = (6,7,8,9,10,11,14,19,97) AND WHYTO = (1,2) THEN 04 Else IF WHYFROM in (1,2) or WHYTO in (1,2) then 04 ELSE = 05 IF TRIPPURP=-9 and WHYFROM in (1,2) or WHYTO in (1,2) then 04	Description: General purpose of trip. Range: -9=Not ascertained 01= Home-based work (HBW) 02= Home-based shopping (HBSHP) 03= Home-based social/recreational (HBSOC) 04= Home-based other (HBO) 05= Not a home-based trip (NHB)
TRPMILES	Description:

Variable and Creation Instructions	Short Description and Ranges
Creation: If LOOPTRIP1 in (1,2,3,97) then TRPMILES=LOOPDISTANCE_MILES; Else TRPMILES=destination_Distance;	Trip distance in miles, derived from route geometry returned by Google Maps API, or from reported loop-trip distance. Range: 0-4859.4769818
URBAN Creation: IF [URBAN AREA TYPE] = "URBANIZED AREA" THEN "01" IF [URBAN AREA TYPE] = "URBAN CLUSTER" THEN "02" IF [GEOMETRY] SURROUNDED BY ([GEOMETRY] WHERE [URBAN AREA TYPE] = "URBANIZED AREA") THEN "03" ELSE "04"	Description: Household's urban area classification, based on home address and 2020 TIGER/Line Shapefile. Range: 01=In an urban area 02=In an urban cluster 03=In an area surrounded by urban areas 04=Not in urban area
URBANSIZE Creation: IF [POPULATION OF URBAN AREA] BETWEEN 50,000 AND 199,999 THEN 01 IF [POPULATION OF URBAN AREA] BETWEEN 200,000 AND 499,999 THEN 02 IF [POPULATION OF URBAN AREA] BETWEEN 500,000 AND 999,999 THEN 03 IF [POPULATION OF URBAN AREA] >= 1,000,000 THEN IF RAIL = "01" THEN 04 IF RAIL = "02" THEN 05 ELSE 06	Description: Urban area size where home address is located. Range: 01=50,000-199,999 02= 200,000-499,999 03= 500,000-999,999 04= 1,000,000 or more with heavy rail 05= 1,000,000 or more without heavy rail 06= Not in an urbanized area
URBRUR Creation:	Description: Household in urban/rural area.

Variable and Creation Instructions	Short Description and Ranges
IF URBAN = (01,02) THEN URBRUR=01 ELSE URBRUR=02	Range: 01=Urban 02= Rural
USEPUBTR Creation: If COUNT(TRPTRANS) where TRPTRANS = (8, 10, 11, 12) then 01; Else 02;	Description: Public Transit Usage on Travel Date. Range: 01= Yes 02= No
VEHAGE Creation: IF VEHYEAR = (-8,-7) THEN VEHYEAR IF VEHYEAR = (2021, 2022, 2023) THEN 1 ELSE 2022-VEHYEAR IF VYEAR < 0 THEN VEHAGE = VEHYEAR	Description: Age of vehicle, based on model year. Range: 1-40
VEHCASEID Creation: ON THE VEHICLE FILE: VEHCASEID equals a concatenation of HOUSEID VEHID. ON THE TRIP FILE: If TRIP_VEHID > 0 then VEHCASEID equals a concatenation of HOUSEID VEHID; Otherwise VEHCASEID=-1.	Description: Unique vehicle identifier. Range: 900001300201-900021804001 -1= Valid skip (TRIP file only)
VMT_MILE Creation: Trip distance in miles for personally driven vehicle trips"	Description: Trip distance in miles for driver trips, derived from route geometry returned by Google Maps API.

Variable and Creation Instructions	Short Description and Ranges
if DRVR_FLG='01'; If DRVR_FLG=01 and TRPTRANS in (01,02,03,04,06,07) then VMT_Mile=TRPMILES; Else VMT_MILE=-1;	Range: 0- 1682.699192 -1 = Valid skip
WEEKEND Creation: If NTSAWAY < 0 then WEEKEND=NTSAWAY Else if NTSAWAY=1 and Day Of Week(LD5) in (6,7) then WEEKEND='01'; Else if NTSAWAY=2 and DOW(LD5) in (5,6,7) then WEEKEND='01'; Else if NTSAWAY=3 and DOW(LD5) in (4,5,6,7) then WEEKEND='02'; Else if NTSAWAY=4 and DOW(LD5) in (3,4,5,6,7) then WEEKEND='02'; Else if NTSAWAY=5 and DOW(LD5) in (2,3,4,5,6,7) then WEEKEND='02'; Else WEEKEND='03';	Description: Trip includes weekend
WHYFROM Creation: If TRIPID ne 1, set WHYFROM equal to WHYTO from the preceding trip (e.g., WHYFROM for TRIPID=3 gets set to WHYTO from TRIPID=2); If TRIPID=1 then do; If FRSTHM=02 then WHYFROM=B1B; else WHYFROM=1; If WHYFROM in (-1,-9) then do; If origin_locname='HOME' then WHYFROM=1; Else if origin_locname='WORK' then WHYFROM=2;	Description: Reason for previous trip. Range: 01= Regular activities at home 02= Work from home (paid) 03= Work at a non-home location 04= Work activity to drop-off/pickup someone/something 05= Other work-related activities 06= Attend school as a student

Variable and Creation Instructions	Short Description and Ranges
Else if origin_locname='SCHOOL' then WHYFROM=6; Else If origin type could be determined (hotel, friend's house, etc.) assign WHYFROM as appropriate; End; If WHYFROM=-1 then WHYFROM=-9;	07= Attend childcare or adult care 08= Volunteer activities (not paid) 09= Change type of transportation 10= Drop off/pick up someone (personal) 11= Health care visit 12= Buy meals 13= Shop/buy/pick-up or return goods 14= Other family/personal errands 15= Recreational activities 16= Exercise 17= Visit friends or relatives 18= Rest or relaxation/vacation 19= Religious or other community activities 97= Something else (specify) -9= Not ascertained
WHYTRP1S Creation: IF WHYTO = (1,2) THEN WHYTRP1S= 01 IF WHYTO = (3,4,5) THEN WHYTRP1S=10 IF WHYTO = (6,7,19) THEN WHYTRP1S= 20 IF WHYTO = 11 THEN WHYTRP1S= 30 IF WHYTO = (13,14) THEN WHYTRP1S= 40 IF WHYTO = (15,16,17,18) THEN WHYTRP1S= 50 IF WHYTO = 10 THEN WHYTRP1S=70 IF WHYTO = 12 THEN WHYTRP1S=80 ELSE WHYTRP1S= 97	Description: Trip purpose summary. Range: 01=Home 10=Work 20=School/Daycare/Religious 30=Medical/Dental services 40=Shopping/Errands 50=Social/Recreational 70=Transport someone 80=Meals 97=Something else

Variable and Creation Instructions	Short Description and Ranges
WHYTRP90 Creation: [HOME TOUR WINDOW] = MINIMUM(TDTRPNUM) WHERE WHYTO IN (1, 2) BETWEEN MAXIMUM(TDTRPNUM) WHERE WHYTO IN (1, 2) [WORK TOUR WINDOW] = MINIMUM(TDTRPNUM) WHERE WHYTO IN (3) BETWEEN MAXIMUM(TDTRPNUM) WHERE WHYTO IN (3) [WHYTRP90 STEP ONE] = IF WHYTO IN (1,2) AND [HOME TOUR WINDOW] = 0 THEN WHYFROM IF WHYTO IN (1,2) AND [HOME TOUR WINDOW] >= 1 THEN WHYTO WHERE DWELLTIME = MAXIMUM(DWELLTIME) OF [HOME TOUR WINDOW] IF WHYTO IN (3) AND [WORK TOUR WINDOW] = 0 THEN WHYFROM IF WHYTO IN (3) AND [WORK TOUR WINDOW] >= 1 THEN WHYTO WHERE DWELLTIME = MAXIMUM(DWELLTIME) OF [WORK TOUR WINDOW] WHYTRP90 = IF [WHYTRP90 STEP ONE] = (3) THEN 01 IF [WHYTRP90 STEP ONE] = (4,5) THEN 02 IF [WHYTRP90 STEP ONE] = (13) THEN 03 IF [WHYTRP90 STEP ONE] = (8,10,14) THEN 04 IF [WHYTRP90 STEP ONE] = (7) AND R_AGE >= 18 THEN 4 IF [WHYTRP90 STEP ONE] = (12) AND [WORK TOUR WINDOW] = (0,1) THEN 04 IF [WHYTRP90 STEP ONE] = (13) AND IN [WORK TOUR WINDOW] THEN 04 IF [WHYTRP90 STEP ONE] = (6,19) THEN 05	Description: Travel day trip purpose consistent with 1990 NPTS design. Range: 01=To/From Work 02=Work-Related Business 03=Shopping 04=Other Family/Personal Business 05=School/Church 06=Medical/Dental 08=Visit Friends/Relatives 10=Other Social/Recreational 11=Other 99=Refused / Don't Know

Variable and Creation Instructions	Short Description and Ranges
IF [WHYTRP90 STEP ONE] = (7) AND R_AGE < 18 THEN 05 IF [WHYTRP90 STEP ONE] = (11) THEN 06 IF [WHYTRP90 STEP ONE] = (17) THEN 08 IF [WHYTRP90 STEP ONE] = (15,16,18) THEN 10 IF [WHYTRP90 STEP ONE] = (12) AND IN [HOME TOUR WINDOW OR NOT WORK TOUR WINDOW] THEN 10 IF [WHYTRP90 STEP ONE] = (97) THEN 11 IF [WHYTRP90 STEP ONE] = (-8,-7) THEN 99 ELSE 11	
WORKER Creation: If 0 =< R_AGE<16 then WORKER=-1; Else do; If PAYPROF=1 or PRMACT=1 then WORKER=01; Else WORKER=02; end;	Description: Employment status of respondent. Range: 01= Worker 02= Not worker -1= Valid skip
WRKCOUNT Creation: Count of records on person file where WORKER= 01, within each household.	Description: Count of workers in household. Range: 0-6

Claritas Variables

Description: Tract and Block Group Variables (also known as Claritas Variables). These variables were added to describe the characteristics of the areas where the NHTS respondents were surveyed. This allows the data analyst to look for patterns in travel behavior, not only by individual characteristics, but also by neighborhood characteristics. The data user can examine how characteristics such as population density,

housing density, renter occupancy rate, and urbanicity of the household location may affect individual travel behavior. These variables are referred to as the “Claritas” variables as the source of the data is Claritas, LLC. (Claritas). This is the fourth NHTS to have these Claritas data appended to enhance analysis, the first three being the 2001, 2009 and 2017 NHTS.

Logic:

Sources of Tract and Block Group Variables: The data contained in these variables were derived from U.S. Decennial Census and American Community Survey data, enhanced by Claritas using data from regional and city planning agencies, federal agencies (e.g., Bureau of Labor Statistics, Bureau of Census, Bureau of Economic Analysis) U.S. Postal Service, the direct mail industry, the real estate industry, and experts in the fields of geographic information systems and mapmaking. These estimates are made at relatively small units of geography, such as census tracts and block groups, which make this update effective for use in supplementing the NHTS data based on the home location. The vintage for the Claritas source data is 2023.

Variable Naming Scheme: The variable names were designed so that:

- many of these variables would fall together in an alphabetic listing, and
- the variable name would help in describing the contents.

The naming scheme is:

First letter	H for household descriptor
	W for workplace descriptor
	S for school location descriptor
	O for trip origin descriptor
	D for trip destination descriptor
Second letter	B for block group level data
	T for tract level data
Third letter of Household variables	H for housing characteristic
	P for population characteristic
Last five letters	Describe the data in the variable, e.g. POPDN = population density; RESDN = residential density

For example, HTHRESDN is a household descriptor, at the tract level, describing a housing characteristic, specifically, residential density (RESDN).

The set of tract and block group variables derived by Claritas are:

Household Descriptor, Block Group Level

HBHRES DN	Housing units per square mile
HBHTNR NT	Percent renter-occupied housing
HBHUR	Urban/rural code (see below)
HBPP OP DN	Population density (persons per square mile)

Household Descriptor, Tract Level

HTEEMP DN	Employed persons per square mile (i.e., employed persons at their residence location)
HTHRES DN	Housing units per square mile
HTHTNR NT	Percent renter-occupied housing
HTPP OP DN	Population density (persons per square mile)

Urban-Rural Continuum: One of the most popular Claritas variables is a variable that describes a location based on where it falls on an urban-rural continuum that is categorized according to five levels of “urbanicity.” This variable is HBHUR and represents urbanicity at the home location block group. This breakout of urban/rural should not be confused with the variable URBAN, which is the Census defined urban area status of the sample household.

The categories of the Claritas Urban/Rural Continuum and the distribution of NHTS households within these categories are presented in the following table, showing the distribution of the weighted and unweighted (number of respondent households) across the urban/rural continuum.